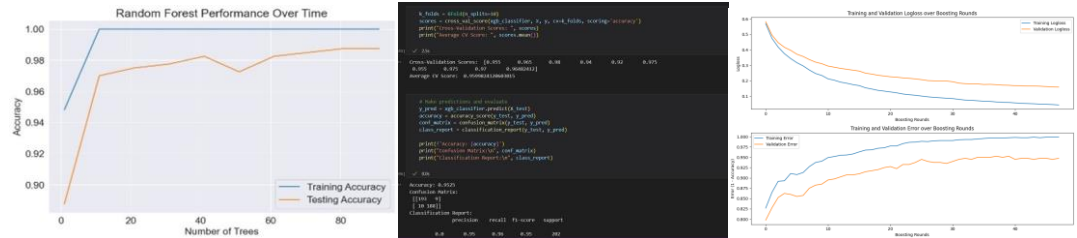


Project Title	Dysarthria Detection Using Machine Learning Techniques		
Project Code	CSCI417	Course Name	Machine Intelligence
Professor	Dr. Ghada Khoriba and Dr Ensaf Hussein		
TA	Eng. Mina, Aly and Aya	Mentor Name	Dr. Ghada Khoriba and Dr Ensaf Hussein
Team Name	Ctrl + shift + Elite		
Team Members	Islam Ahmed Nabil	Hossam Eldeen Anwer Ahmed	Hazem Nasr Radwan
	Mahy Hesham Abdallah	Farida Ahmed Ismail	
Problem Summary	Identifying and classifying dysarthria—a speech disorder caused by impaired muscle control—remains a manual and subjective process. Existing methods lack efficiency and struggle with diverse dysarthria types. This project aims to automate dysarthria assessment using advanced machine learning models. Our goal is to create a precise, scalable, and technology-driven solution for identifying dysarthric speech.		
Methodology	The audio files recorded during clinical sessions, each associated with individual data such as gender, were organized in a CSV file, with entries indicating the gender and corresponding audio file names for each session participant. • Exploratory Data Analysis: A comparison between visualizations was made for four random samples (Male, Female, dysarthric, non-dysarthric) like waveplot, spectrogram, Spectral Rolloff and zero crossing rate to inspect the difference between the dysarthric and the non-dysarthric. • Data Augmentation: Although the dataset was originally augmented, two types of data augmentation were applied to the audio files, noise injection and time shifting to add noise to both classes to be less prone for audio noise misclassification, and time shifting to detect whether the person is dysarthric or not from any start time. • Feature Extraction: In the final preprocessing stage, the audio signals were transformed for a more structured format. Mel-frequency cepstral coefficients (MFCC) were employed to extract 39 numerical coefficients from the frequency domain of each audio signal, optimizing the dataset for model training. • Training: Lastly, 5 different models were implemented (SVM, Random Forest, Logistic Regression, RNN, XGBoost). Meanwhile, hyperparameter tuning was performed to identify the optimal parameters for each model.		
Achievements and Skills Gained	1- Gained skills in preprocessing speech data. 2- Gaining skills in the extraction of critical features from speech data, such as Mel Frequency Cepstral Coefficients (MFCCs) and spectrograms. 3- Explored and implemented strategies for optimizing machine learning models, enhancing accuracy, and reducing prediction errors.		

Project Title	Dysarthria Detection Using Machine Learning Techniques
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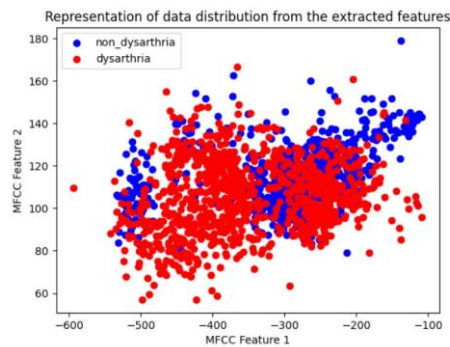
(Cont.)

Main Results



The Random Forest model performs well on audio files due to its ensemble approach, combining multiple decision trees. This model handled the non-linear relationship in the data. The XGBoost model effectively classified dysarthria with a 0.96 cross-validation score and 0.9525 accuracy, indicating its strong capability in detecting speech variations. The RBF kernel of the support vector machine has achieved an accuracy of 97.75% while the linear kernel achieved 91% which indicates the data is not linearly separable. The RBF kernel has achieved an average cross validation score of 96.79%

Discussion and Conclusion



After assessing the five models on +90% accuracy because of the non-linear nature of the MFCC features. The image of reduced MFCC dimensions shows how a RBF kernel of SVM can perform better than the linear one. while a logistic regression model can struggle with that kind of data distribution. In summary, SVM, XGBoost, and Random Forest proved their efficiency for non-linear MFCC data.

References

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Future Work and Suggestions

Develop a user-friendly interface to be a diagnostic tool, ensuring accessibility for healthcare professionals, caregivers, and individuals seeking self-assessment to explore the integration of real-time dysarthria detection, enabling immediate feedback for individuals, and facilitating on-the-spot assessment.

Group Photo

